

NEXORA AUTHENTIC PREVIOUS YEAR QUESTIONS

Exam: UPSC

Subject: Forestry

Questions: 15

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AI-generated PYQs: 0

1. [UNKNOWN] paper-i

1. Justify that the study of silvics is essential for successful afforestation programme in India. 8 Explain different kinds of thinning and its application in forest. 8 What is sustained yield? Explain how it is achieved in practice. 8 Define the characteristics of mangrove forests. 8 Discuss the physiology of root parasitism in sandal tree. 8

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2. [UNKNOWN] paper-i

2. Explain the ecophysiological factors that are more concerned to silviculturist. 15 Enlist the classification of silvicultural systems on the basis of mode of regeneration and pattern of felling. 15 How is the rotation of any particular species at any particular locality practically decided? 10

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3. [UNKNOWN] paper-i

3. (a) What is a mother tree? How is a plant prepared through vegetative propagation different from a plant raised through seed? 15 (b) (i) Write the phenology of the following : Tectona grandis Melia dubia Shorea robusta Cedrus deodara (ii) Can climate-change change the period of phenology? Share with examples. 10 (c) Write the problems and prospects of exotic tree species in India with suitable examples. 15 FSI-P-FSTY/45 2 ===== PAGE 3 =====

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4. [UNKNOWN] paper-i

4. (a) What is conversion? Discuss the need of conversion of one silvicultural system to another. 15 Give the silvicultural characteristics and economic importance of the following : 10 Azadirachta indica (11) Acacia catechu (0 Enlist different types of nurseries and write different types of nursery beds used in a nursery. (ii) Enlist different types of containers used in a forest nursery and explain different methods of seed sowing followed in a nursery. 15 SECTION—B

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5. [UNKNOWN] paper-i

5. Write in brief on the criteria of selection of tree for resistance to adverse environments for high quality timber production. What are the measures to be taken into consideration during preparation of earthen check dam in the forest areas? 8 Write in detail about the influence of parent rock in the distribution of tree species. 8 Explain the environmental and economic role of community forestry in India. 8 Write the salient features of the Solid Waste Management Act, 2000 and 2016. What new initiatives have been taken in the Solid Waste Management Rules, 2016? 8

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6. [UNKNOWN] paper-i

6. Write about the diagnostic and design survey of agroforestry. Can it help the farmers in the integration of tree with crops to enhance the crop productivity in agroforestry system? Justify. 10 Write about the pre- and post-Environmental Impact Assessment (EIA) of any mining area of India. Does GIS help in EIA? Write the name of the software used in Environmental Impact Assessment for the mining areas. 10 Explain seed production and certification system in Indian forestry. 10 What is the importance of heritability and how can genetic gain be estimated in tree improvement programme? 10 FSI-P-FSTY/ 45 3 [P.T.O. ===== PAGE 4 =====

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7. [UNKNOWN] paper-i

7. (a) Write the soil-water relationship of any forest area. Describe the influence of water table in the growth and

development of tree species. 10 Describe incomplete mating designs used in tree improvement. 10 What are the causes of forest fire? What measures are taken to protect forests against damage by fire? 10 What are the different selection methods used by the tree breeders? 10

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8. [UNKNOWN] paper-i

8. (a) What is hydrology? Describe the role of hydrology in planning and management of watershed development. Does tree species improve the infiltration rate, soil temperature, water level and hydrological cycle? Justify with few examples. 10 What is sustainable development? Write about the criteria and indicator of sustainability fulfilling the needs and demands of growing population of India. 10 What is farm forestry? Write about the objectives, difficulties and financial return from the farm forestry. 10 How would you develop tree improvement programmes for raising productivity in forestry? 10 * * * FSI-P-FSTY/45 4 BS9-918

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9. [UNKNOWN] paper-ii

Q2. (a) What are the precautions required for diameter measurements with calipers ? Discuss the errors that occur due to non-observation of the precautions. 15 What is growing stock? How is normal growing stock calculated in clear felling system based on final MAT? 15 Define rotation. Explain different types of rotation with special reference to ecological, industrial and economical benefits. 10

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10. [UNKNOWN] paper-ii

Q3. (a) Define working plan. Discuss objectives and scope of a working plan. Describe salient features of a good working plan. 15 What is increment ? Discuss different types of increments. Discuss the graphical relationship between current annual increment and mean annual increment. 15 Define volume tables and give their classification. 10 FSI-D-FSTY 2 ===== PAGE 3 =====

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11. [UNKNOWN] paper-ii

Q4. (a) Define forest management. Give its objectives. How does the attitude of the owner put impact on the management of forests? 15 Define forest sampling. Give advantages of sampling Discuss different types of non-random sampling methods used in forestry. 15 Define remote sensing. Discuss its application in forest management along with GIS applications. 10 FSI-D-FSTY 3 ===== PAGE 4 ===== SECTION B

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12. [UNKNOWN] paper-ii

Q5. What are the industrial uses of gums and resins ? Discuss the factors affecting the production and supply of gums and resins. 8 What is carbon sink ? How do forest soils act as important carbon sinks? 8 Forest ecology is the study of complex interactions between organic and inorganic components of the forest ecosystem. Explain organic and inorganic components providing two examples of each as they relate to the forest ecosystem. 8 What is the ecological and economical importance of biodiversity ? Mention the salient measures for conservation of biodiversity. 8 Write down four advantages of timber seasoning and suggest which one is commercially suitable over other methods, with suitable examples. 8

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13. [UNKNOWN] paper-ii

Q6. List out different types of preservatives used for protection of timber against fungi and insects and classify them based on solvent used. 15 How is enumeration of animal population in natural forests carried out in general ? Also specify the method(s) adopted in the case of tigers. 15 What is succession and climax? Give the causes of forest succession. 10

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14. [UNKNOWN] paper-ii

Q7. Define ethnobotany. Write in detail about the role of ethnobotany in modern medicine and its approaches to the drug industry. 15 What are the objectives of Joint Forest Management (JFM). Give methods used for preserving forest resources through JFM. 15 Enlist the common and scientific names of trees and shrubs (seven each) having medicinal importance in ethnobotany. 10

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15. [UNKNOWN] paper-ii

Q8. What are the key elements of the wood industry and paper industry strategies ? Explain. 15 What are the main differences between forest policy and forest laws ? Give salient points of the National Forest Policy of 1952 and 1988. 15 Describe the defects that appear during seasoning in timbers. 10 FSI-D-FSTY 4

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